Design and Evaluation of a Database-Centric Agentic Architecture for Nursing Home Evacuation Coordination in Storm Surge Scenarios

Master's thesis defence

Anik Dewanje

Technische Universität Ilmenau, Databases and Information Systems Group
Responsible professor: Prof. Dr.-Ing. habil. Kai-Uwe Sattler Supervisor: M. Sc. Saba Zamankhani

Moving a nursing home before the water arrives

- A storm surge is forecast for Hamburg. People near the Elbe have to leave in time.
- Nursing home residents need a planned operation: buses, staff and medical care that the home does not own.
- Eight organizations take part. Each sees only its own part, and all work against one deadline.
- The plan rarely stays as it was. A road closes, a shelter reports fewer beds, a resident collapses on a moving bus.

Scripted simulation (DV 100)
One branch for each disruption the author expected. Robust on anticipated exceptions only (Reichert and Weber 2012; Woods 2015).

LLM agents
Observe, reason and act in a loop, and revise the plan (Yao et al. 2023). Could they handle a disruption nobody scripted?

Case from the RESCUE-MATE project (Research for Civil Security), written at Absolute Software GmbH, Hamburg.

Research questions

A design-and-evaluation study (Hevner et al. 2004): build the artifact, evaluate it against a scripted baseline, report the answer either way.

RQ1 Can a database-centric agentic architecture coordinate a nursing home evacuation better than a scripted workflow baseline?

RQ2 If it cannot, why not? Where does the deficit come from?

RQ3 What would have to change, and what should a designer consider, when building an agentic simulation for emergency coordination?

The boundary of RQ1

None of our scenarios required adaptation. A controller that ignores every disruption scores the same as the baseline. We answer RQ1 as "did the agentic architecture win here", and make no claim about whether it can ever win.

Related work and the gap

Work	What the LLM agents play	Compared with a non-LLM controller?
RESPOND (Sultimov et al. 2026)	The population in a flood	No, a what-if platform demo
Lee et al. (2025)	Evacuees with personas	No, predicts real decisions
Li et al. (2026)	Design study with practitioners	No, more useful for procedures than prediction
TravelPlanner, TheAgentCompany (Xie et al. 2024; Xu et al. 2025)	Task agents, scored by code	No, models against models
Relief optimization (Huang et al. 2012; Yin et al. 2024)	No agents, one planner who sees all	Not applicable
This thesis	**The eight coordinating organizations**	**Yes, a DV 100 script, 310 runs**

To our knowledge, no prior work puts LLM agents in the role of the coordinating organizations and scores them against a doctrine-grounded script.

The case: one small evacuation, tight on purpose

- Augustinum: **100 residents**, water at 12:00
- **2 buses** $\times$ **50 seats**, bus-2 is the reserve
- **2 schools** $\times$ **60 beds**, so 20 spare beds
- **10 agents** in **8 organizations**, 6 of them with an LLM

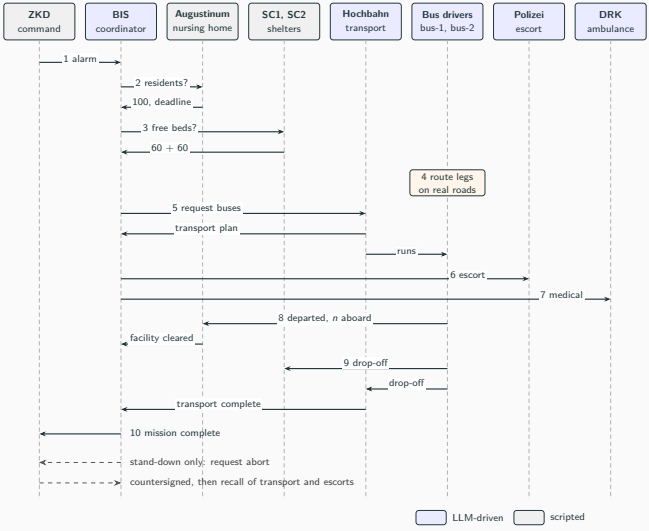
No bus carries everyone in one trip, and no shelter takes everyone.

ID	Scenario
Anticipated arm	
S01	Closed loop, nothing goes wrong
S02	Road closure
Novel arm, written after the baseline was frozen	
S04	Medical emergency in flight
S05	Unverified persons report, later withdrawn
S06	Conflicting evacuation orders
S07	Reduced bus capacity (bus-1 may carry 25)
S08	Partial shelter capacity loss (SC1 reported at 40)

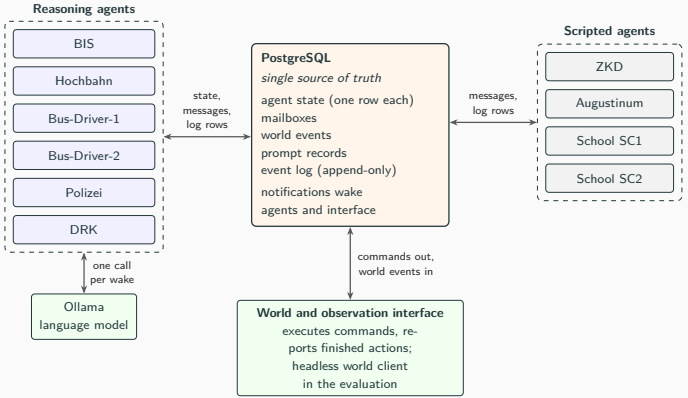
True of every scenario

Every disruption arrives as a message and leaves the simulation world unchanged. A plan that ignores it still saves everyone.

Who talks to whom in one run

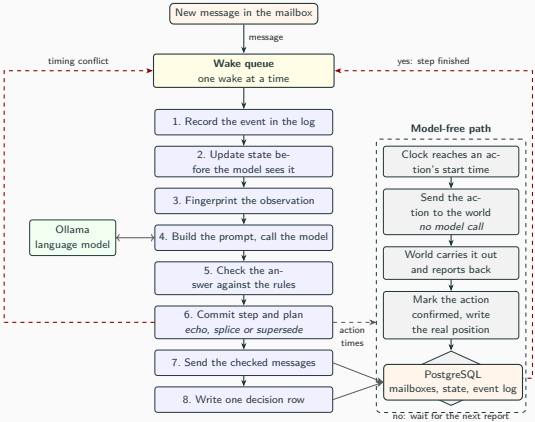


Agentic system: ten agents, one database



- Agents never message each other directly.
- PostgreSQL holds all state, carries every message and wakes the recipient.
- Only the six reasoning agents call the model.
- Stack: Ollama, a custom asyncio engine, PostgreSQL.

One wake of an agent, and how it replans



The model proposes a step. The runtime owns time, validation and the audit row. A new step meets the committed one in one of four ways:

Echo Same step again. Nothing changes.

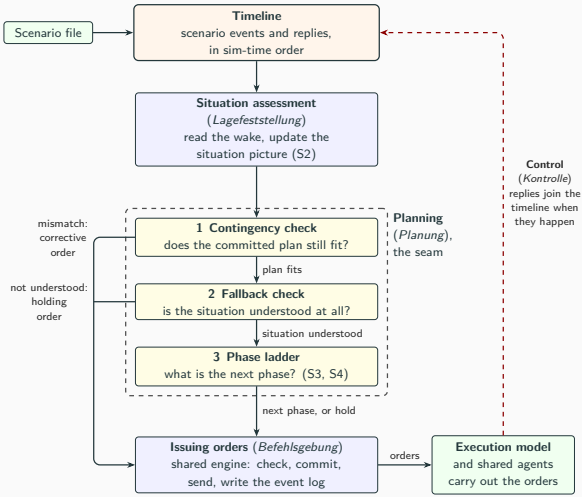
Splice Sent actions stay, only the waiting tail is replaced.

Supersede A sent action changes. The old step is aborted and the new one starts over.

Back-pressure Timing conflicts stop after 3 attempts or when the gap stops shrinking.

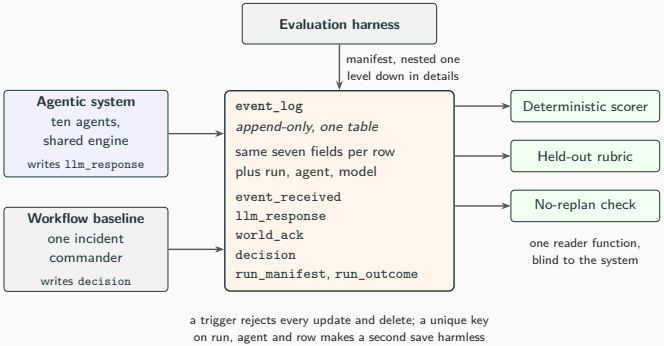
Temperature 0.2. No retry of malformed output.

Workflow baseline: the DV 100 command cycle



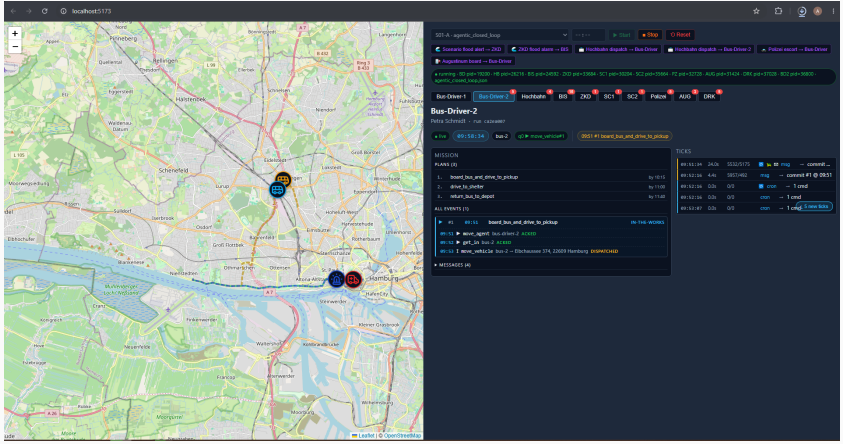
- One scripted Incident Commander replaces BIS and Hochbahn. The other eight agents are shared.
- No language model in the loop.
- Four phases, three anticipated branches, one generic hold.
- At the point where an agent would call its model, three checks run in a fixed order: correct, hold, advance.
- A simulator carries out the orders, where the agentic side runs live field agents.

Data architecture: one log for both systems



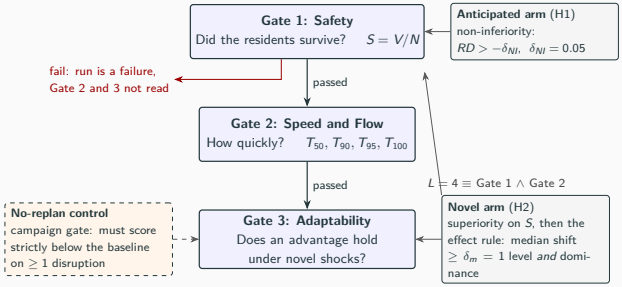
- 12 tables: 5 runtime, 6 sensing, 1 audit.
- `event_log` is append-only. A database trigger rejects every update and delete.
- Both systems write the same rows. One deterministic scorer reads them.
- 310 scored runs wrote 83,661 rows.

Observation interface



We can watch a live agentic run on the map, and open the full prompt and answer behind any single decision.

Evaluation: three gates, read in order

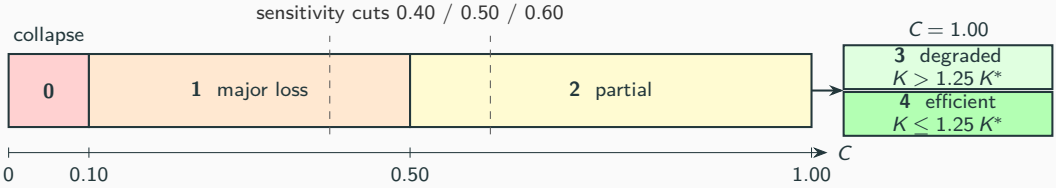


- 7 scenarios, $k = 10$ paired runs per cell, common random numbers.
- 310 scored runs. Three models completed the novel arm: glm-5.2, deepseek-v4-pro, qwen3.5.
- Pre-registered: $\delta_{NI} = 0.05$ on survival, $\delta_m = 1$ whole level.
- Four hypotheses. H1 and H2 are confirmatory.
- The third model slot changed after data were seen, so the novel-arm model set is exploratory.

What we measure

$$S = \frac{\text{valid residents}}{\text{ceiling}}$$

Valid means delivered before the deadline, into a shelter within its declared beds. The ceiling is 100.



Every cut has a reason written before the runs: 0.10 is one part-load of one bus, 0.50 is half the home. Level 4 also needs the run to drive at most 25 % further than the same system and model's undisrupted runs.

RQ1: both confirmatory hypotheses fail

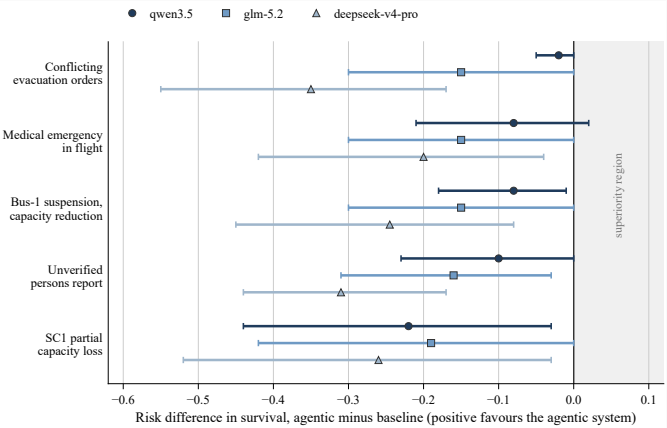
Mean survival on the 50 novel runs per system

System / model	mean S
workflow baseline	**1.000**
agentic / qwen3.5	0.894
agentic / glm-5.2	0.840
agentic / deepseek-v4-pro	0.727

qwen3.5 capped at 1.0 per run (three runs scored 1.1).

- **Gate 1 fails 26 of 27 cells.** Three are on the road closure, which is not like-for-like. Without them, 23 of 24.
- **H1** (non-inferiority, anticipated arm): holds in 1 of 8 cells, glm-5.2 on the undisrupted `closed_loop`.
- **H2** (superiority, novel arm): 0 of 19 cells pass, and no cell reaches the one-level minimum effect.
- H3 is unresolved. H4 was not tested: every run used one time scale.

The baseline wins on all five novel disruptions



Every point estimate is below zero, and no cell shows superiority. The gap shrinks with the model: mean risk difference -0.27 (deepseek), -0.16 (glm-5.2), -0.10 (qwen3.5).

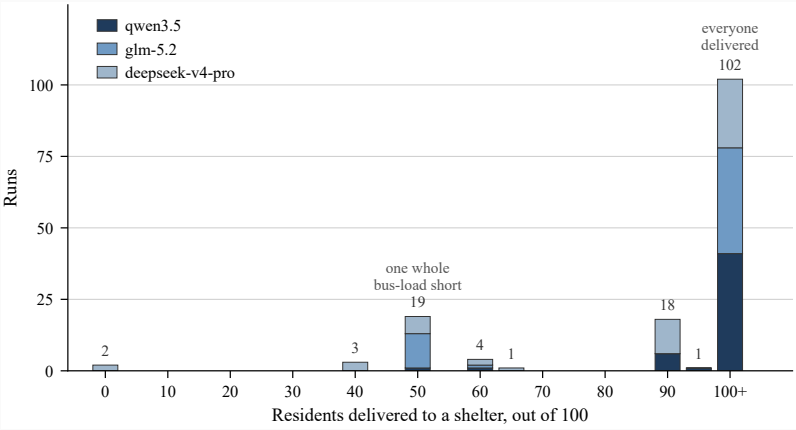
RQ2: what we ruled out

Tested against the 150 agentic novel runs. These tests were designed after the data were seen, so they are exploratory.

Candidate cause	Verdict
The agents were told the wrong population	No. 100 residents reported in 150 of 150 runs.
The database message layer failed	No. 93.8 % of 1,932 addressed requests were answered.
Too few missions dispatched, reserve bus held back	No. About 2.3 dispatch orders in failing and successful runs alike. Bus-2 used in all 150 runs.
The scorer removed real deliveries	No. De-duplicated arrivals track sheltered residents.
Drivers over-claimed their headcount	Does not explain it. An over-report raises S.
The run ran out of clock	Mostly no. 5 of the 19 total-loss runs dropped off after 12:00.

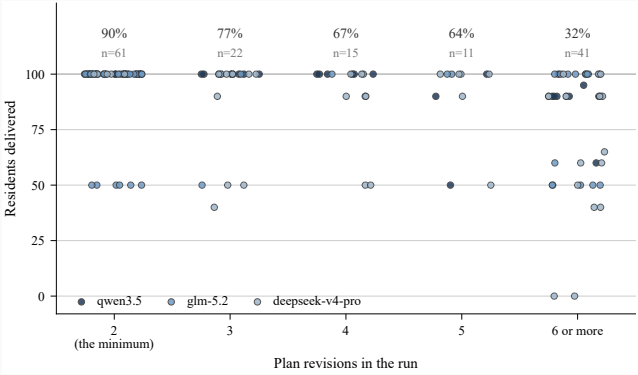
What remains is that legs already dispatched were not carried through to a drop-off.

The loss comes in discrete amounts



Of 150 novel runs, 102 delivered everyone, 19 delivered 50 (one bus-load short) and 18 delivered 90. A slow loss of quality would spread the runs out.

More plan revisions, fewer residents delivered

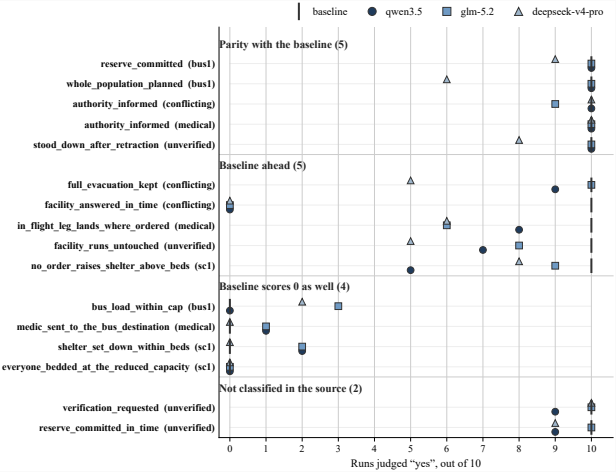


Full delivery falls from **90 %** at the fewest revisions to **32 %** at six or more.

Leading hypothesis: a commitment failure

- Comprehension was not measured.
- Causal direction is open: 57 of 108 aborts are stand-downs after a failure.
- Live field agents against a simulator (deviation 7): we cannot separate command post from field.

Decision quality is split



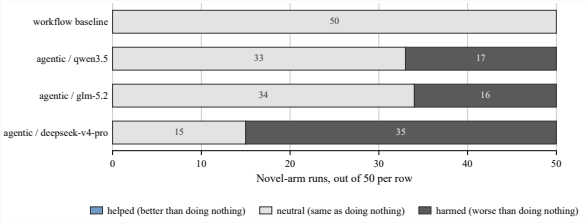
16 held-out rubric criteria, checked on the event log and fixed before the novel runs.

- **Parity on 5**, all about deciding: inform the authority, commit the reserve, plan for everyone.
- **Baseline ahead on 5**, mostly about closing the loop.
- `facility_answered_in_time`: baseline 10 of 10, every agentic model 0.
- On 4 more the baseline also scores 0.

The test could not be lost

A no-replan control ignores every disruption. On all five novel scenarios it commits byte-identical orders, drives the same 60.63 bus-km and reaches the same level as the baseline.

- The baseline's $S = 1.000$ is the null policy's score, so it says nothing about adaptation.
- Gate 0 admits 0 of the 6 disruption scenarios.
- The agentic system is below doing nothing in all 15 novel cells, and in 68 of 150 runs.
- Part of the gap exists undisrupted: `closed_loop` S is 1.000 (glm), 0.950 (qwen3.5), 0.790 (deepseek).



What the agentic approach costs

System / model	messages	msgs / resident	tokens	budget fraction
workflow baseline	13.2	**0.13**	0	0.00
agentic / deepseek-v4-pro	91.9	1.26	725,457	1.80
agentic / qwen3.5	106.6	1.18	894,243	2.28
agentic / glm-5.2	111.8	1.33	829,906	2.70

- Nine to ten times the baseline's messages per resident delivered.
- 140 of 150 agentic novel runs finished over the time budget.
- The baseline is deterministic and calls no model.

RQ3: what a designer should consider

Consideration	Measured in this study
1. Commit, then confirm. A leg on the road stays committed unless the agent states a reason, and the runtime counts it. Every leg gets an explicit delivery confirmation.	Full delivery 90 % $\rightarrow$ 32 % as plan revisions rise
2. Make the problem real. A disruption must change the state the scorer reads. SC1 was reported at 40 beds and still declared 60.	Orders byte-identical to the no-replan control
3. Test the test first. Run Gate 0 before the campaign, size the design before the runs, fix and pin the models.	Sign test cannot go below $p = 0.0625$ at 5 scenarios
4. Plan the cost. Decide what the extra messages and tokens have to buy.	9 to 10 times the messages per resident

These come from one evaluation of one architecture on one case. The commitment boundary rests on an association, not yet on a cause.

Contributions

1. **An agentic architecture and its baseline.** Ten agents coordinating through PostgreSQL, and a DV 100 baseline that shares the scenarios, actions and scoring.
2. **A scored comparison with a negative result.** 310 runs, seven scenarios, pre-registered hypotheses, deterministic scoring.
3. **A diagnosis of the loss.** Messages worked. Dispatched legs were not carried through. Leading hypothesis: commitment.
4. **A method for a fair comparison.** A shared append-only event log, regret against a null policy, and Gate 0 as a check on treatment validity.

Conclusion

Answers

RQ1 On our scenarios, no. The baseline won on all five novel disruptions, and the agentic system scored below doing nothing. Whether it can ever win remains open.

RQ2 Messages worked and missions were dispatched. Committed legs were not carried through to delivery. Leading hypothesis: a failure of commitment.

RQ3 Commit, then confirm. Make disruptions change the world. Pass Gate 0 first. Plan the cost.

Next experiment

A commitment-constrained variant on the same scenarios and models. If it delivers more, plan churn is a cause. If not, churn is a symptom. Then scenarios that pass Gate 0.

Thank you.

Questions?

Backup: the seven parity deviations

#	Deviation	Effect on a result
1	A baseline message's id is its position in the run, not a clock time.	None. No code reads it.
2	A baseline message names its receiver.	None. No code reads it.
3	A baseline record is labelled as a decision.	None. Scored facts do not change.
4	A baseline record has an empty model name.	None. The scorer reads no model name.
5	One node, the IC, holds the command and the fleet picture.	Favours the baseline. This is the doctrine.
6	The IC plans its schedule with road distance and travel time.	Favours the baseline.
7	Only the baseline's scenario files declare a world. A simulator carries out the IC's orders; the agentic side runs its field agents live.	Mixed. No scripted drop-offs: against the baseline. Simulator in place of live agents: for it.

Backup: risk differences per novel cell

$RD = S_{\text{agentic}} - S_{\text{baseline}}$, paired, with 95 % CI.

Disruption	qwen3.5	glm-5.2	deepseek-v4-pro
Conflicting evacuation orders	−0.02 [−0.05, 0.00]	−0.15 [−0.30, 0.00]	−0.35 [−0.55, −0.17]
Medical emergency in flight	−0.08 [−0.21, 0.02]	−0.15 [−0.30, 0.00]	−0.20 [−0.42, −0.04]
Reduced bus capacity	−0.08 [−0.18, −0.01]	−0.15 [−0.30, 0.00]	−0.245 [−0.45, −0.08]
Unverified persons report	−0.10 [−0.23, 0.00]	−0.16 [−0.31, −0.03]	−0.31 [−0.44, −0.17]
SC1 partial capacity loss	−0.22 [−0.44, −0.03]	−0.19 [−0.42, 0.00]	−0.26 [−0.52, −0.03]

Backup: anticipated arm

System / model	`closed_loop`	`road_closure`
workflow baseline	1.000	0.60 against the nominal 100
agentic / glm-5.2	1.000 (H1 pass)	0.84
agentic / qwen3.5	0.950	not run
agentic / deepseek-v4-pro	0.790	0.70

On the road closure the baseline's file makes one school unreachable and declares 60 residents, while the agentic file has 100 and a detour. The two systems ran different scenarios there.

Backup: Gate 0 evidence

Scenario	world	orders identical	level L (C) doctrine	level L (C) no-replan	bus-km doctrine	bus-km no-replan	discri-minates
fixture: `shelter_closure`	yes	no	3 (1.0)	0 (0.0)	83.12	38.14	**yes**
`medical_emergency_in_flight`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`unverified_persons_report`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`conflicting_evacuation_orders`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`bus1_suspension_capacity_reduction`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`sc1_partial_capacity_loss`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`closed_loop`	yes	yes	3 (1.0)	3 (1.0)	60.63	60.63	no
`road_closure`	no	no	3 (1.0)	3 (1.0)	83.12	38.14	no

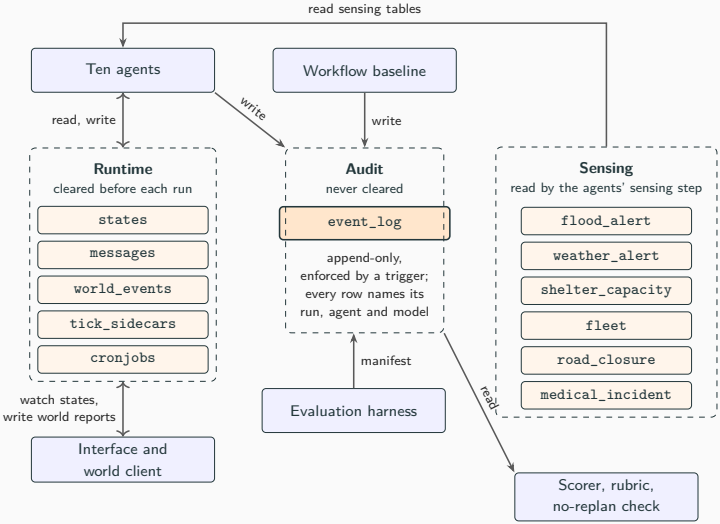
Level capped at 3 here. Only the shelter-closure fixture discriminates.

Backup: why the registered H2 primary could not decide

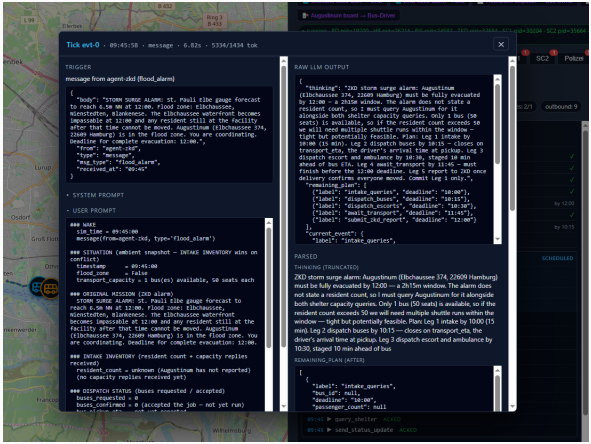
- Registered primary: a Clopper–Pearson interval on $P(L_{\text{agentic}} \geq L_{\text{baseline}})$ per cell, a sign test across scenarios, Holm–Bonferroni. Implemented after the data were seen (Amendment 15).
- It agrees with the risk difference: no cell is above chance.
- At $k = 10$ every interval is about 0.6 wide.
- A two-sided exact sign test over five scenarios cannot return less than $p = 0.0625$. With two ties the floor is 0.250, and qwen3.5 and glm-5.2 sit on it.

The risk difference, continuous and paired, carries the argument.

Backup: the twelve tables

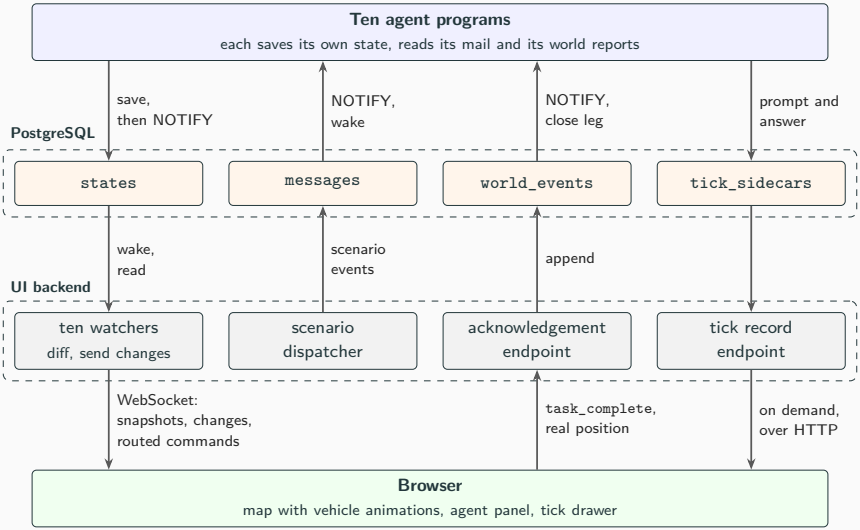


Backup: inside one decision



BIS woken by the storm surge alarm: the prompt on the left, the raw model output and the parsed plan on the right.

Backup: how the interface reads the database



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